



HK IMO Training 2021-2022
Problem Set 12
For 1 January 2022



Problem 45. Let $\mathbb{N}$ be the set of all positive integers. Find all $f : \mathbb{N} \rightarrow \mathbb{N}$ such that

$$(n-1)^2 < f(n)f(f(n)) < n^2 + n$$

for every positive integer n .

Problem 46. Find all positive integers n satisfying the following condition: It is possible to partition the set $\{1, 2, \dots, 2n\}$ into n pairs of numbers, such that for each pair (a, b) , the number $ab + 1$ is a perfect square.

Problem 47. Let $ABCD$ be a circumscribed quadrilateral. Let W, X, Y, Z be the tangential points of the incircle of $ABCD$ with AB, BC, CD, DA respectively. Let E, F, G, H be the midpoints of WX, XY, YZ, ZW respectively. Prove that AF, BG, CH, DE enclose a cyclic quadrilateral.

Problem 48. Let m and n be positive integers, and let S be a finite set of some m -tuples of integers. Consider a game played between two players as follows. Initially, there are m piles of coins put in a row with a fixed order, with the first pile having n coins, and the other piles being empty. The two players take turns choosing an m -tuple $(x_1, x_2, \dots, x_m)$ from S , and add x_j coins to the j th pile for each $j = 1, 2, \dots, m$ (if x_j is negative, adding x_j coins to a pile means removing $-x_j$ coins from that pile, given that the pile has at least $-x_j$ coins). The player who cannot make a valid move will lose the game. Prove that there exist m and S such that the first player has a winning strategy if n is a power of 2, and the second player has a winning strategy if n is not a power of 2.